

Lab 4: Combinational Circuits

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Introduction

This laboratory explores the fundamental design and application of combinational logic circuits used for comparisons and data management. The experiments transition from magnitude comparators, which evaluate the value of binary numbers, all the way to decoders and multiplexers, which serve as important mechanisms for address translation and signal routing. By implementing these circuits, it can be demonstrated to show how simple logic gates can be scaled into complex systems and how a decoder's ability to identify a binary state is utilized by a multiplexer to select and transport data across a digital bus.

Summary

The lab involved designing and implementing combinational logic circuits using hierarchical design techniques in Quartus. The experiment began with constructing a one-bit magnitude comparator and verifying its operation through truth tables and simulation results. The design was then expanded to create a two-bit magnitude comparator capable of comparing two binary numbers and determining whether one value is greater than, equal to, or less than the other.

A 2-to-4 decoder circuit was also implemented to convert binary input combinations into specific output lines. Finally, a 4-to-1 multiplexer was designed to demonstrate how one of several input signals can be selected and routed to a single output using control signals. All circuits were verified through simulation and hardware testing on the Altera DE2-115 development board to confirm their correct logical operation.

Background

Combinational circuits are an important part of digital systems because their outputs depend only on the current inputs. These circuits are used to perform logical operations such as comparison, decoding, and data selection. Devices such as comparators, decoders, and multiplexers are commonly used in processors and digital control systems.

A magnitude comparator is used to compare two binary numbers and determine whether one value is greater than, equal to, or less than the other. Decoders convert binary input combinations into specific output lines, while multiplexers select one of several input signals and route it to a single output based on select inputs.

These combinational circuits demonstrate how basic logic gates can be combined to perform useful digital operations and form the building blocks of larger digital systems.

Objective of the Report

The main objective of this experiment was to design, simulate, and implement several combinational circuits using Quartus and FPGA hardware. The specific goals included:

- Designing and verifying a 1-bit magnitude comparator
- Expanding the design to implement a 2-bit magnitude comparator
- Constructing and testing a 2-to-4 decoder circuit
- Designing and implementing a 4-to-1 multiplexer
- Verifying circuit functionality through simulation waveforms
- Implementing the circuits on the Altera DE2-115 FPGA board
- Comparing theoretical results with simulation and hardware outputs

Equipment

Hardware

- **Altera DE2-115 Development and Education Board**
 - Used to implement and test magnitude comparators, 2-4 decoders and 4-1 multiplexers.
 - Toggle switches used to dictate inputs in each part of the lab.
 - LEDs used to display outputs.

USB-Blaster Programming Cable

Used to program the FPGA device on the DE2-115 board

Software:

- **Intel Quartus Prime / Quartus II Graphic Editor**
 - Used to create circuits for 1 and 2 bit magnitude comparators.
 - Used to create circuits for 2 - 4 decoders.
 - Used to create circuits for 4 -1 multiplexers.

Quartus Simulation Waveform Editor (.vwf files)

- Used to verify functional correctness of theoretical circuits designed in the pre-labs.
- Used to compare theoretical results with simulated outputs

Logic Components:

- XNOR gates

- AND gates
- OR gates
- Not gates

Input and Output Devices:

- Switches for inputs.
- LEDs for outputs.

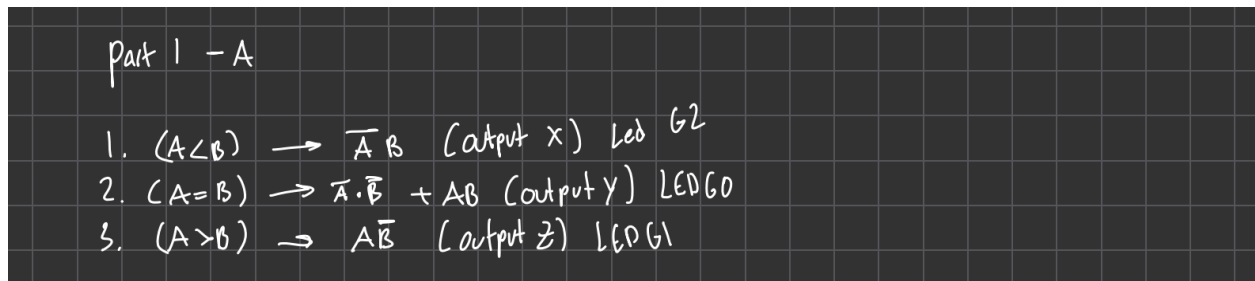
Additional Equipment:

- Laboratory computer capable of running Quartus
- USB cable for board connection

Pre-Lab

Part 1 - A

The objective of the pre-lab was to analyze the comparative logic and right simplified logic forms which include and/or gates. This helps to correctly form a logic circuit that compares two inputs of 0 and 1.



This is a very simple 1 bit Magnitude Comparator. Output X holds the value of A being less than B, and works by an and2 gate consisting of notA with B. Output Y holds the value for A equaling B, and functions in an or2 gate consisting of the outputs of Anot and Bnot into one input with the and gate of A and B as the other input. Output Z holds the value for A being larger than B which works opposite to the output X where instead it is an and gate of inputs A and Bnot. Truth tables shown were later used to verify the correctness of the waveforms.

Part 1 - B

Part 1-B

$$1. (A < B) \rightarrow = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0 \quad (\text{Output } x) \quad \text{LEDD2} = x$$

$$2. (A = B) \rightarrow = \bar{A}_1 A_0 B_1 B_0 + \bar{A}_1 A_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0 + A_1 A_0 B_1 B_0 \rightarrow (A_1 \oplus B_1) (A_0 \oplus B_0) \quad \text{LEDD1} = y$$

$$3. A > B \rightarrow = A_1 \bar{B}_1 + A_1 A_0 \bar{B}_0 + A_0 \bar{B}_1 B_0 \quad (\text{Output } z) \quad \text{LEDD0} = z$$

This has the exact same purpose with corresponding outputs as the previous logic circuit in part 1, however it now compares two bits as opposed to 1. This gives priority to A and B inputs that are labeled 1 over the ones labeled with 0. This is because A1 holds the second bit which if it is high and A0 is low it would mean the number being shown is actually 2 instead of 1. This takes this into account and compares the inputs resulting in outputs x, y, z which hold the same exact compare function as the previous part. Truth tables shown were later used to verify the correctness of the waveforms.

Part 2

Part 2:

(expressions)

$$Y_0 = \bar{D}_1 \cdot \bar{D}_0$$

$$Y_1 = \bar{D}_1 \cdot D_0$$

$$Y_2 = D_1 \cdot \bar{D}_0$$

$$Y_3 = D_1 \cdot D_0$$

Truth table:

D ₁	D ₀	Y ₀	Y ₁	Y ₂	Y ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

This shows the logic and truth table for a 2-4 decoder. A decoder works by reading the inputs D1 and D0 and activating 1 output at a time that corresponds to such inputs. For example D1 and D0 being 0 and 1 respectively means that Y1 would be high. The circuit is composed of two not gates and 4 two-input and gates. Truth tables shown were later used to verify the correctness of the waveforms.

Part 3

Part 3) 1) • 4 data inputs
 • 2 select inputs ∴ 4-1 MUX
 • One output Y

2) $Y = \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3$

3) $S_1 = 0$
 $S_0 = 1 \rightarrow Y = D_1$

4)

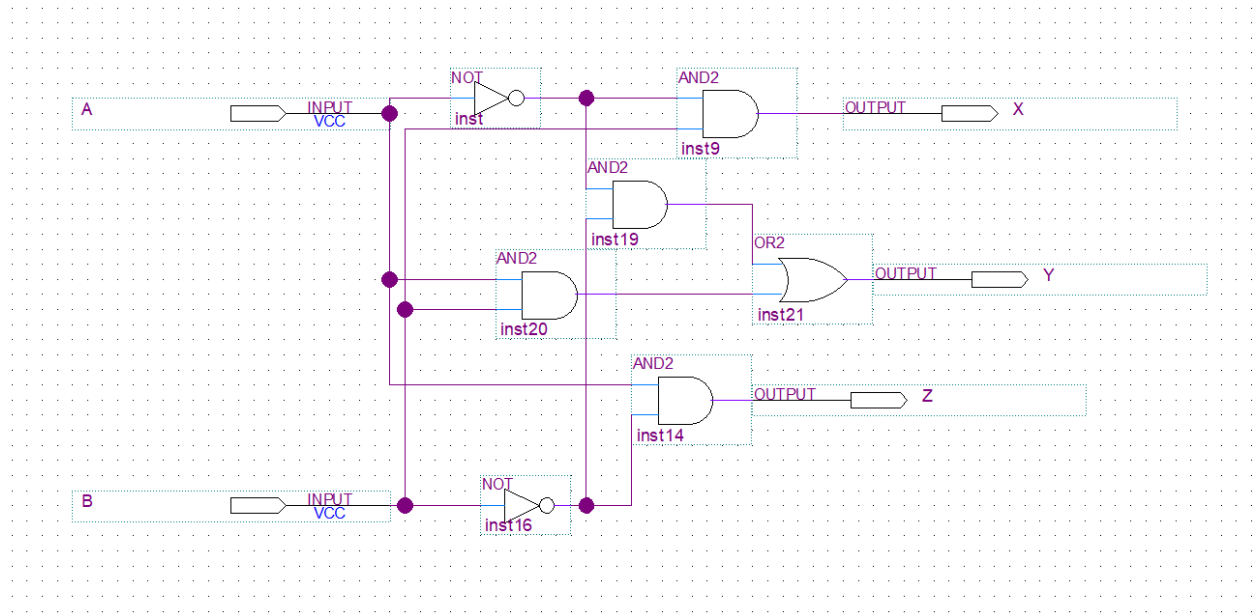
S_1	S_0	D_0	D_1	D_2	D_3	Y
0	0	0	x	x	x	0
0	0	1	x	x	x	1
0	1	x	0	x	x	0
0	1	x	1	x	x	1
1	0	x	x	0	x	0
1	0	x	x	1	x	1
1	1	x	x	x	0	0
1	1	x	x	x	1	1

A 4-1 multiplexer functions as a digital selector that routes one of four data inputs to a single output based on the binary state of two select lines. Internally, the circuit utilizes a 2-to-4 decoder structure to generate four mutually exclusive minterms, each of which is ANDed with its corresponding data bit. The final OR gate then combines these signals, ensuring that only the data from the addressed input appears at the output terminal Y. The logic works by essentially representing a unique condition where a specific data bit is allowed to pass. For example the term $S_1 \text{not} * S_0 * D_1$ acts like a gate that only opens when $S_1=0$ and $S_0 = 1$, the value of D_1 then is passed through to the final or gate to match that. Truth tables shown were later used to verify the correctness of the waveforms.

Design Procedure

Part 1 - A

Part 1 - A consisted of designing a magnitude comparator that serves the function of comparing two inputs A and B and returning outputs xyz like explained in the pre-lab.



Uses Two Inputs:

- A
- B

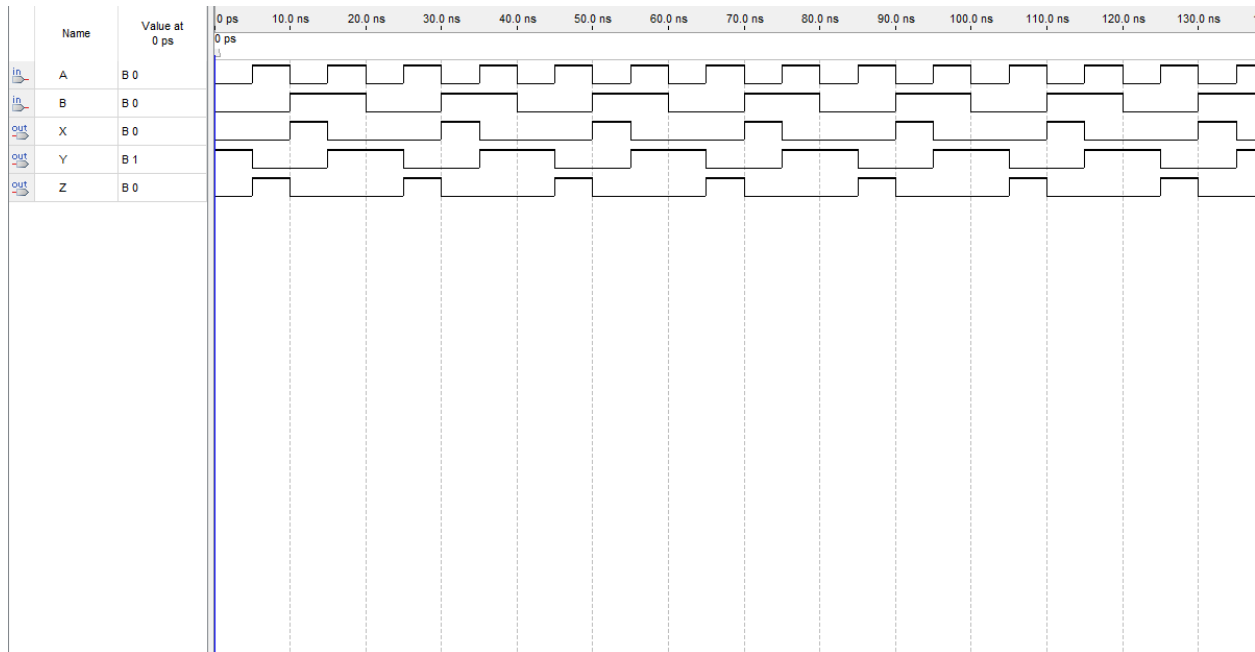
Outputs:

- X ($A < B$)
- Y ($A = B$)
- Z ($A > B$)

Boolean Expressions Used:

- X: $A'B$
- Y: $A'B' + AB$
- Z: AB'

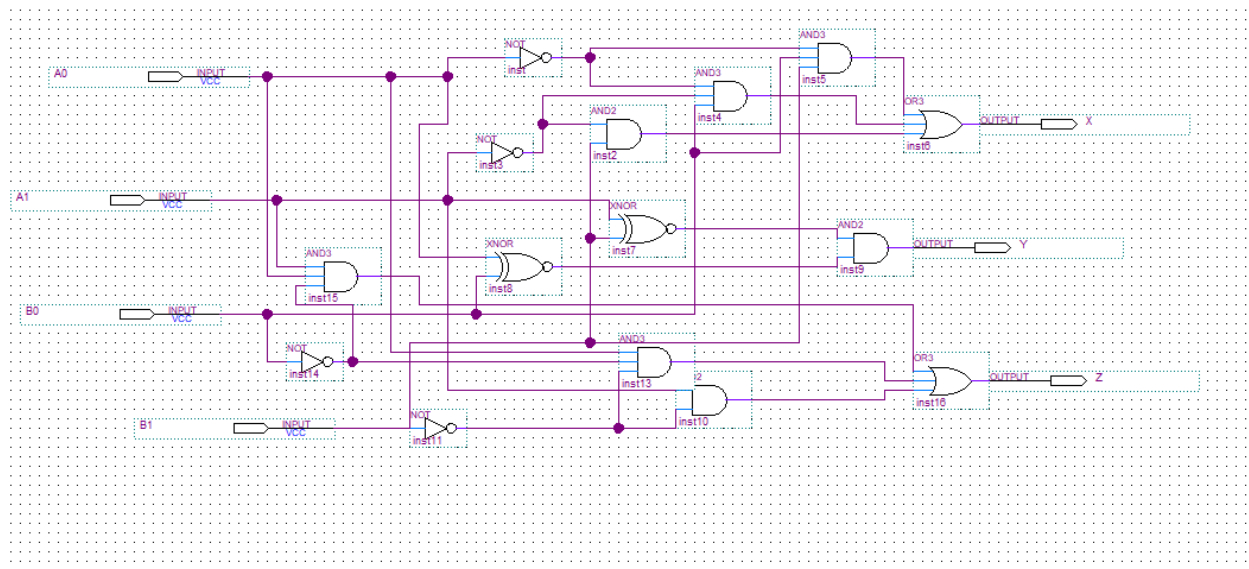
Simulation Waveform:



Overall the simulation results matched exactly with the truth table and no errors were found to interfere with the simulation process. Pins were then assigned and demonstrated for observing behavior on the Altera board.

Part 1 - B

Serves the same function as Part 1 - A but with two bit inputs instead one.



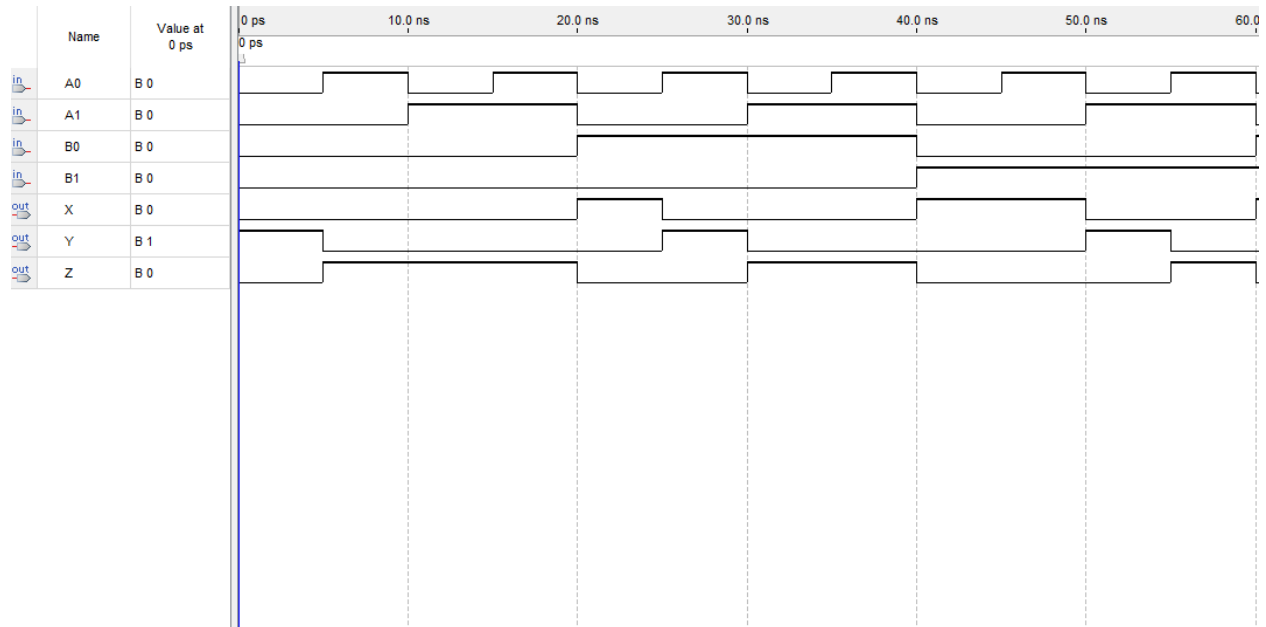
Inputs Used:

- A1 and A0 (MSB and LSB respectively).
- B1 and B0 (MSB and LSB respectively).

Outputs Used:

- X: (A<B)
- Y: (A=B)
- Z: (A>B)

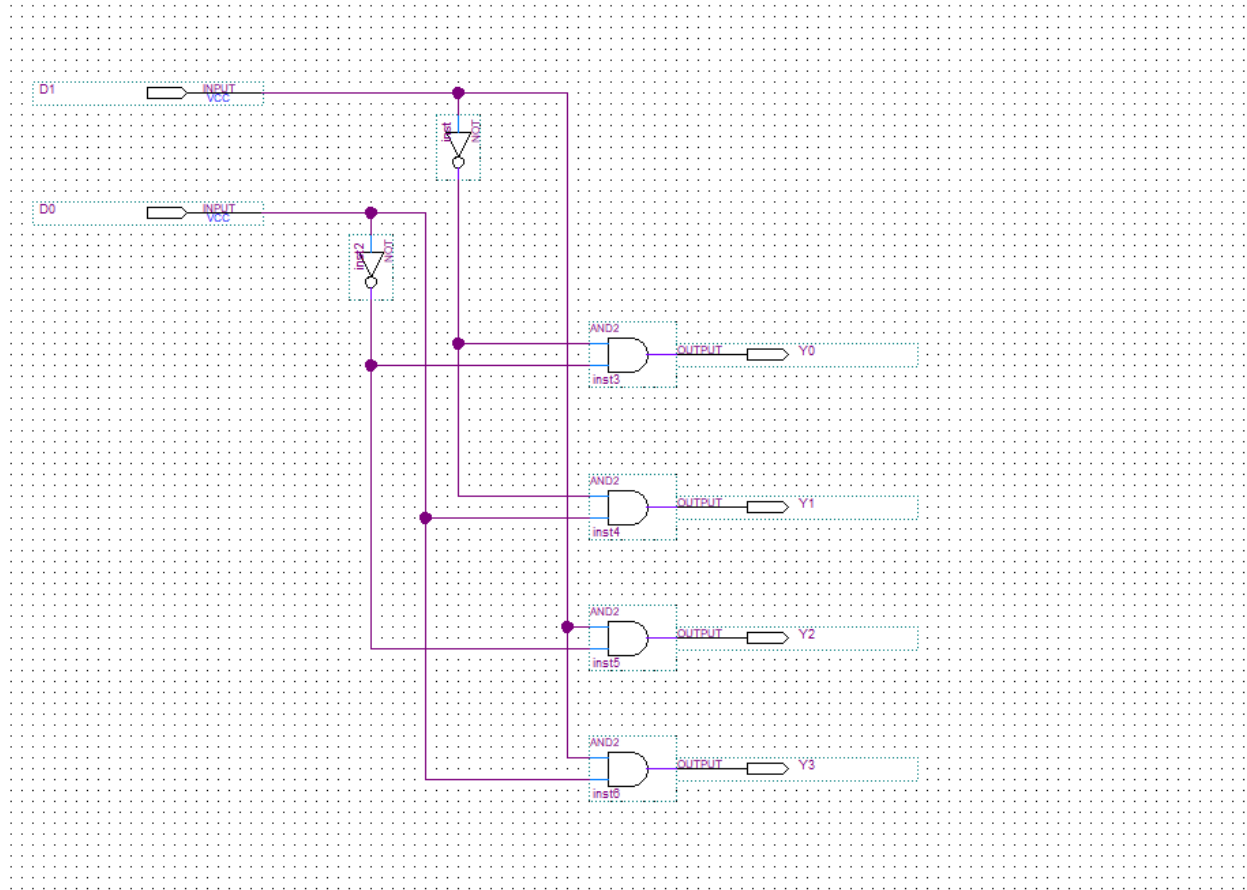
Simulation Waveform:



The simulation waveform shows that it is true and accurately matches the theoretical truth table. In cases where both A and B have at least one input that is high the one with the high input being on the most significant bit is the one that is larger. For example A1: 1 A0: 0 vs B1: 0 B0 : 1, it's comparing 10 (2 in decimal) vs 01 (1 in decimal) resulting in output Z to be high. Pins were then assigned and demonstrated for observing behavior on the Altera board.

Part 2

This circuit is designed to function as a 2-4 decoder which takes inputs and outputs the Y variable corresponding to the number the input holds.



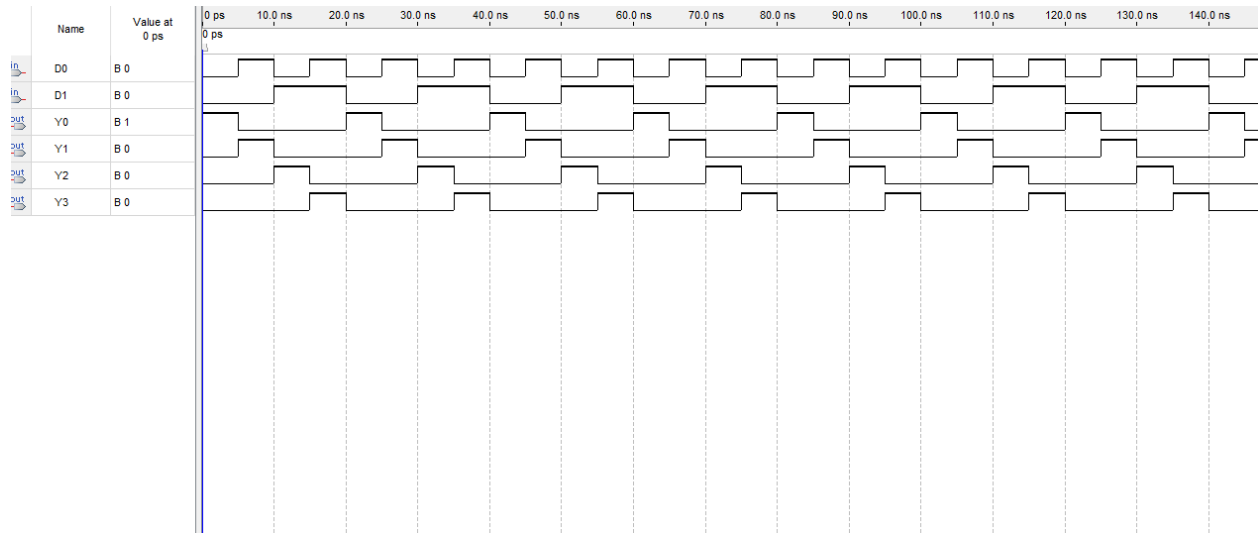
Inputs Used:

- D1
- D2

Outputs:

- Y0 → corresponding to 00 (D1 = 0 , D0 = 0)
- Y1 → corresponding to 01
- Y2 → corresponding to 10
- Y3 → corresponding to 11

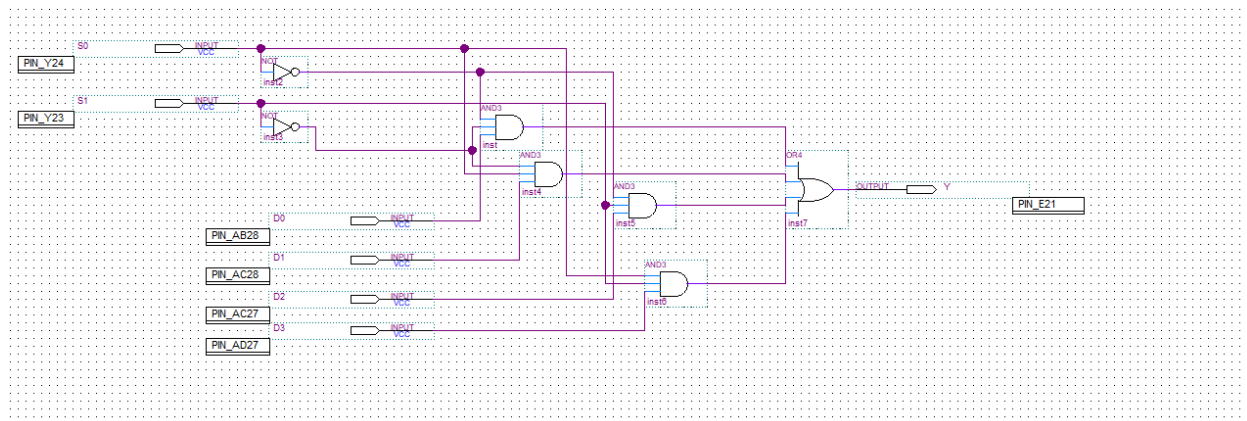
Simulation Waveform:



The outputs shown from the waveform accurately confirmed and matched the theoretical truth table from the pre-lab. Pins were then assigned and demonstrated for observing behavior on the Altera board.

Part 3

This circuit functions as a digital switch that selects one of four data inputs D0 through D3 and forwards it to a single output Y. Essentially it is a 4-1 multiplexer.



Inputs Used:

- 2 Select Inputs:

S0

S1

- 4 Data Inputs:

D0

D1

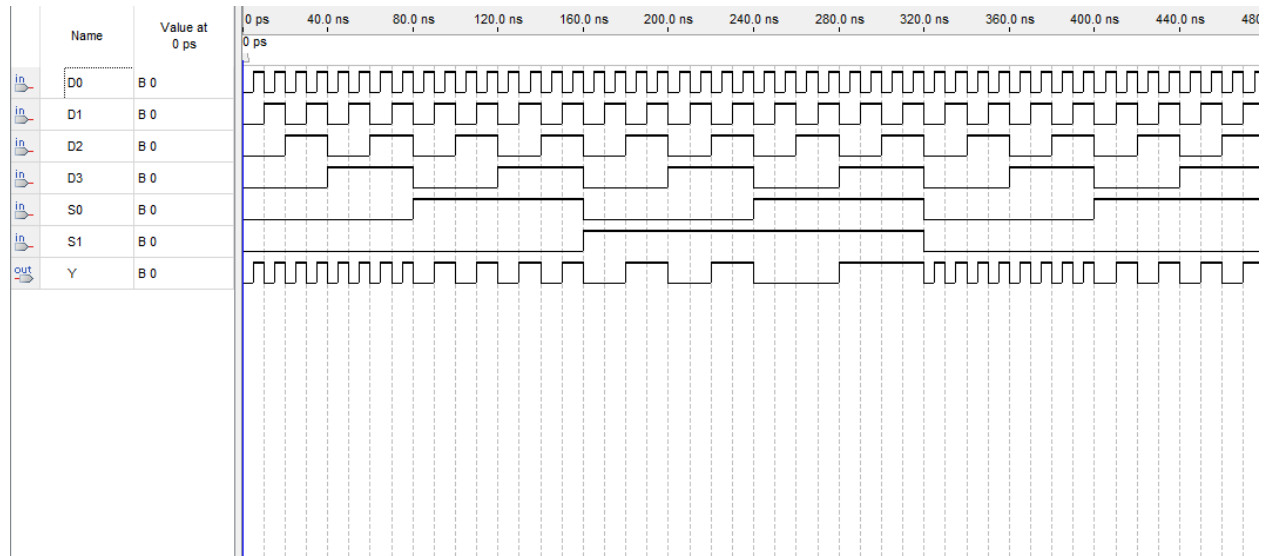
D2

D3

Boolean Expression:

$$Y = S1'S0'D0 + S1'S0D1 + S1S0'D2 + S1S0D3$$

Simulation Waveform:



The waveform matched the experimental truth table and all results were as expected. Pins were then assigned and demonstrated for observing behavior on the Altera board.

Discussion

The results obtained from both simulation and hardware implementation confirmed the correct operation of the combinational circuits designed in this lab. Each circuit behaved according to the theoretical logic derived during the preparation stage.

For the 1-bit magnitude comparator, the simulation verified that the outputs correctly indicated the relationship between the two inputs A and B. When the inputs were equal, the equality output was activated. When A was greater than B, the greater-than output was asserted, and when A was smaller than B, the less-than output was activated. The observed outputs matched the truth table provided in the lab instructions.

The 2-bit magnitude comparator extended this concept by comparing two binary numbers composed of two bits each. The circuit successfully evaluated the significance of the most significant bits first and then the least significant bits when necessary. Simulation results showed that the outputs accurately represented whether A was greater than, equal to, or less than B for all input combinations.

The 2-to-4 decoder circuit correctly translated the two input bits into one of four active outputs. At any given time, only one output line was active while the remaining outputs remained inactive. This confirmed the correct logical implementation of the decoder circuit, where each output corresponds to a specific binary input combination.

The 4-to-1 multiplexer functioned as expected by selecting one of four input signals and forwarding it to the output based on the values of the select lines. During simulation, changing the select inputs caused the output to switch between the different input signals accordingly. This behavior demonstrated the multiplexer's ability to control data flow in digital systems.

Conclusion

This lab investigated the design and implementation of several fundamental combinational circuits using Quartus and hardware. The circuits implemented included a 1-bit magnitude comparator, a 2-bit magnitude comparator, a 2-to-4 decoder, and a 4-to-1 multiplexer.

Simulation results confirmed that each circuit produced outputs consistent with the expected theoretical logic and truth tables. The magnitude comparators correctly determined the relationships between binary numbers, the decoder successfully activated the appropriate output line based on input values, and the multiplexer properly selected input signals according to the select lines.

Hardware testing on the Altera DE2-115 development board further validated the designs. The circuits operated when implemented, produced outputs identical to those predicted during simulation.

Overall, this lab reinforced the principles of combinational logic design and demonstrated how digital circuits can be constructed from basic logic gates to perform useful functions such as comparison, decoding, and signal selection.

Appendices

Appendix A - Part 1 - A (MAGNITUDE COMPARATOR)

- Part1_A.bdf
- Part1_A.qsf
- Part1_A.qws
- Part1_A.qpf
- Waveform.vwf

Appendix B - Part 1 - B (2 BIT MAGNITUDE COMPARATOR)

- Part1_B.bdf
- Part1_B.qsf
- Part1_B.qpf
- Part1_B.qws
- Waveform.vwf

Appendix C - Part 2 (2 - 4 DECODER)

- Part2.qsf
- Part2.qpf
- Part2.bdf
- Part2.qws
- Waveform.vwf

Appendix D - Part 3 (4 - 1 MUX)

- Part3.bdf
- Part3.qpf
- Part3.qsf
- Part3.qws
- Waveform.vwf